

Q1. Cowper's glands are characterized by:

- a. Provides 65% - 75% of seminal fluid
- b. Their duct joins the membranous part of the urethra
- c. They neutralize any traces of acidic urine
- d. Their secretion is rich in fibrinolysin

Answer: c. They neutralize any traces of acidic urine

Explanation: Cowper's (Bulbo-urethral) glands produce clear mucus secretion that serves to lubricate the penile urethra and neutralize any traces of acidic urine before the passage of semen.

- a. WRONG - It is the seminal vesicles that give 70-80% of seminal fluid, not Cowper's glands.
- b. WRONG - Their duct joins the initial portion of penile urethra, not the membranous part.
- c. CORRECT - Their function is to lubricate penile urethra and neutralize acidic urine traces.
- d. WRONG - It is the prostate whose secretion is rich in fibrinolysin, not Cowper's glands.

[Reference: Dr. Ahmed Zahra's Notes, Page 37, Section: The Bulbo-Urethral (Cowper's) Glands]

Q2. Which of the following characterizes the mature Graafian follicle?

- a. Theca interna cells can differentiate into smooth muscle cells
- b. Gap junctions are present in the perivitelline space
- c. Corona radiata is a single layer of flat cells
- d. Theca externa secrete androgen

Answer: b. Gap junctions are present in the perivitelline space

Explanation: In the mature Graafian follicle, corona radiata cells send cytoplasmic processes that penetrate zona pellucida and contact oocyte microvilli via gap junctions in the perivitelline space.

- a. WRONG - Theca interna cells secrete androgen and do not differentiate into smooth muscle.
- b. CORRECT - Gap junctions sit in the perivitelline space between corona radiata and oocyte.
- c. WRONG - Corona radiata is a single layer of columnar granulosa cells, not flat cells.
- d. WRONG - It is theca interna that secretes androgen; theca externa has no secretory role.

[Reference: Dr. Ahmed Zahra's Notes, Page 47, Section: Mature Graafian (Tertiary) Follicles]

Q3. The blood-testis barrier is formed by:

- a. Tight junctions between Leydig cells
- b. Tight junctions between adjacent Sertoli cells
- c. Basement membrane of seminiferous tubules alone
- d. Peritubular myoid cells

Answer: b. Tight junctions between adjacent Sertoli cells

Explanation: Sertoli cells are linked by occluding tight junctions that split the seminiferous epithelium into basal and adluminal compartments – the blood-testis barrier.

- a. WRONG - Leydig cells lie in the interstitium and do not form the barrier.
- b. CORRECT - Sertoli-Sertoli tight junctions form the barrier.
- c. WRONG - The basement membrane alone is not sufficient to isolate germ cells.
- d. WRONG - Myoid cells assist tubule contraction but do not form the barrier.

[Reference: Dr. Ahmed Zahra's Notes, Page 21, Section: Seminiferous Tubules & Blood-Testis Barrier]

Q4. Which cell type in the anterior pituitary secretes ACTH?

- a. Somatotrophs
- b. Corticotrophs
- c. Thyrotrophs
- d. Gonadotrophs

Answer: b. Corticotrophs

Explanation: Corticotrophs in the pars distalis produce ACTH (adrenocorticotrophic hormone) from proopiomelanocortin.

- a. WRONG - Somatotrophs secrete growth hormone.
- b. CORRECT - Corticotrophs secrete ACTH.
- c. WRONG - Thyrotrophs secrete TSH.
- d. WRONG - Gonadotrophs secrete FSH and LH.

[Reference: Dr. Ahmed Zahra's Notes, Page 88, Section: Anterior Pituitary Cell Types]

Q5. Kupffer cells are:

- a. Sinusoidal endothelial cells
- b. Fixed macrophages of the liver sinusoids
- c. Hepatocytes in zone 1
- d. Vitamin-A storing cells of the space of Disse

Answer: b. Fixed macrophages of the liver sinusoids

Explanation: Kupffer cells are stellate mononuclear-phagocyte-system macrophages resident in hepatic sinusoids; they clear senescent RBCs, endotoxins and microbes.

- a. WRONG - Sinusoidal endothelium is a distinct cell type.
- b. CORRECT - Kupffer cells are the fixed macrophages of the sinusoids.
- c. WRONG - Zone 1 hepatocytes are parenchymal cells, not macrophages.
- d. WRONG - Vitamin-A storage is done by Ito (hepatic stellate) cells.

[Reference: Dr. Ahmed Zahra's Notes, Page 105, Section: The Liver Sinusoids and Kupffer Cells]

Q6. Type I alveolar cells (pneumocytes) are characterized by:

- a. Cuboidal shape with surfactant granules
- b. Simple squamous cells forming the gas-exchange surface
- c. Ciliated columnar epithelium
- d. Goblet cells producing mucin

Answer: b. Simple squamous cells forming the gas-exchange surface

Explanation: Type I pneumocytes are thin squamous cells that cover ~95% of the alveolar surface and are the primary site of gas exchange.

- a. WRONG - Cuboidal cells with surfactant granules describe type II pneumocytes.
- b. CORRECT - Type I are simple squamous, providing the thin gas-exchange surface.
- c. WRONG - Ciliated columnar epithelium lines the airways, not alveoli.
- d. WRONG - Goblet cells are found in bronchi and larger airways.

[Reference: Dr. Ahmed Zahra's Notes, Page 132, Section: Alveolar Epithelium (Pneumocytes Type I and II)]

Q7. The functional unit of the kidney is:

- a. The renal lobule
- b. The nephron
- c. The medullary pyramid
- d. The collecting duct

Answer: b. The nephron

Explanation: The nephron – glomerulus plus its tubule (PCT, loop of Henle, DCT) – is the histophysiological functional unit of the kidney.

- a. WRONG - The lobule is an anatomical subdivision, not the functional unit.
- b. CORRECT - The nephron is the functional unit.
- c. WRONG - The pyramid contains many nephrons but is not itself the unit.
- d. WRONG - Collecting ducts drain many nephrons; they are not the unit.

[Reference: Dr. Ahmed Zahra's Notes, Page 150, Section: The Uriniferous Tubule: Nephron and Collecting Duct]

Q8. Which of the following is characteristic of ceruminous glands?

- a. Merocrine sweat glands
- b. Modified apocrine glands of the external auditory meatus
- c. Sebaceous glands of the eyelid
- d. Endocrine glands of the skin

Answer: b. Modified apocrine glands of the external auditory meatus

Explanation: Ceruminous glands are modified apocrine glands in the external auditory meatus that, together with sebaceous glands, produce cerumen (earwax).

- a. WRONG - Eccrine sweat glands are a different, distributed gland type.
- b. CORRECT - Ceruminous glands are modified apocrine glands of the ear canal.
- c. WRONG - Sebaceous glands of the eyelid are Meibomian glands.
- d. WRONG - Ceruminous glands are exocrine, not endocrine.

[Reference: Dr. Ahmed Zahra's Notes, Page 178, Section: Skin Appendages: Ceruminous and Modified Apocrine Glands]

Q9. Purkinje fibers of the heart are:

- a. Modified skeletal muscle fibers
- b. Modified cardiac muscle fibers specialised for rapid impulse conduction
- c. Autonomic nerve fibers
- d. Smooth muscle bundles

Answer: b. Modified cardiac muscle fibers specialised for rapid impulse conduction

Explanation: Purkinje fibers are large, pale, glycogen-rich modified cardiac myocytes forming the terminal conducting network beneath the endocardium.

- a. WRONG - The heart is not composed of skeletal muscle.
- b. CORRECT - They are modified cardiomyocytes for fast conduction.
- c. WRONG - Autonomic nerves modulate rate but are not Purkinje fibers.
- d. WRONG - Smooth muscle is not the conducting system.

[Reference: Dr. Ahmed Zahra's Notes, Page 66, Section: The Cardiac Conduction System]

Q10. Which cells are responsible for bone resorption?

- a. Osteoblasts
- b. Osteocytes
- c. Osteoclasts
- d. Osteoprogenitor cells

Answer: c. Osteoclasts

Explanation: Osteoclasts are large multinucleated cells of monocyte-macrophage lineage that resorb bone matrix within Howship lacunae.

- a. WRONG - Osteoblasts form new bone matrix.
- b. WRONG - Osteocytes maintain existing matrix; they do not resorb.
- c. CORRECT - Osteoclasts resorb bone.
- d. WRONG - Osteoprogenitor cells differentiate into osteoblasts.

[Reference: Dr. Ahmed Zahra's Notes, Page 58, Section: Bone Cells: Osteoblasts, Osteocytes and Osteoclasts]

Q11. The predominant cell of the germinal center in a lymph node is:

- a. Plasma cells
- b. Centroblasts and centrocytes (activated B cells)
- c. T helper cells
- d. Macrophages

Answer: b. Centroblasts and centrocytes (activated B cells)

Explanation: Germinal centers are sites of B-cell proliferation, somatic hypermutation and class switching; they are dominated by centroblasts (dark zone) and centrocytes (light zone).

- a. WRONG - Plasma cells emerge from germinal centers but reside in medullary cords.
- b. CORRECT - Centroblasts and centrocytes populate the germinal center.
- c. WRONG - T helper cells assist but are not the predominant population.
- d. WRONG - Macrophages phagocytose apoptotic B cells but are outnumbered.

[Reference: Dr. Ahmed Zahra's Notes, Page 199, Section: Lymph Node: Cortex and Germinal Centers]